Part A — Project Proposal

1) App Concept

App Name: Weekly Plate

Problem Statement:

Planning meals for the week is a pain because recipes are all over the place and making a shopping list takes extra time. Weekly Plate lets you pick meals for each day, see the full recipe, and automatically creates a shopping list for the week so cooking at home feels easier and faster.

Target Audience:

Busy **college students** and **working adults** who cook at home and want a simple way to plan meals and shop once for the week.

Key Features:

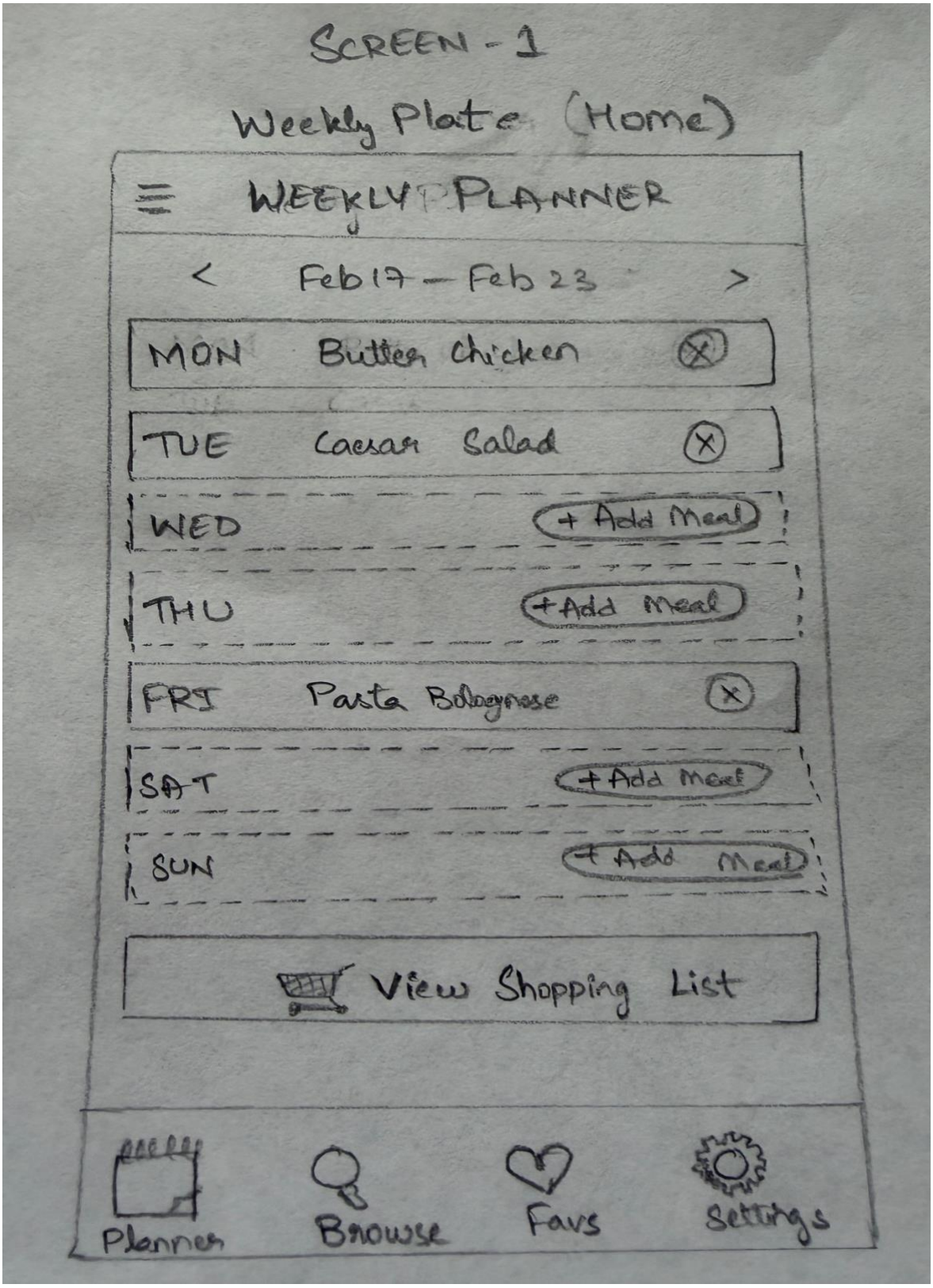
- Browse meals by category (Chicken, Beef, Seafood, etc.)
- Search meals by name
- See recipe details (ingredients + steps)
- Add meals to specific days on a weekly calendar
- Auto-build a shopping list from your planned meals (combines duplicates)
- Mark/unmark recipes as favorites from Recipe Details

Data Source:
Weekly Plate uses the public recipe API **TheMealDB** to pull meal categories, meal lists, search results, and full recipe details.

API Endpoints:

- **List categories:** `categories.php`
- **Browse meals by category:** `filter.php?c=<CategoryName>`
- **Search meals by name:** `search.php?s=<Keyword>`
- **Get full meal details (ingredients + instructions):** `lookup.php?i=<MealId>`

2) Wireframes



SCREEN - 2

Browse Categories.

BROWSE RECIPES

Search Recipes

Categories :

Chicken | Beef | Seafood

Desert | Vegetarian | Lamb

Pork | Pasta | Turkey

+more...

Loading Categories...

Network Error
couldn't load recipes. check your connection
Retry

Planner | Browse | Favs | Settings

SCREEN-3

Recipe List (Grid)

< CHICKEN RECIPES

Search in Chicken...

img
Butter chicken

img
Chicken Tikka Masala

img
BBQ chicken

img
Chicken Alfredo

img
Roast chicken

img
chicken Stew

Planner | Browse | Favs | Settings

SCREEN - 4
Recipe Details

< Butter chicken

Recipe Image (thumbnail)

Chicken | Indian

Ing
chi
Butt
Tom
Cle
Cra
Ing

AUGUST 19

choose a day

Monday ✓

Tuesday

Wednesday

Thursday

Cancel | Confirm

Planner | Browse | Favs | Settings

SCREEN - 5

Shopping List

Shopping List ↑Share

Week : Feb 17 - Feb 23 - 3 meals planned

- [x] Butter — 9 tbsp total
- [ ] Chicken — 1.5kg total
- [ ] Garlic — 6 cloves
- [ ] Pasta — 400g
- [ ] Tomato Sauce — 400ml
- [ ] Cream — 200ml
- [ ] Parmesan — 50g
- [ ] Olive Oil — 4 tbsp

— Empty state (shown when no meals) —

No meals planned for this week

Planner | Browse | Favs | Settings

SCREEN - 6

Settings

Settings

Appearance

Dark Mode off System on

Exclude Categories

These won't appear in Browse

- [ ] Beef
- [x] Seafood (exclude)
- [ ] lamb
- [ ] Pork
- [ ] chicken

Preferences saved locally

Planner Browse Fave Settings

Navigation Flow

- **Planner → Browse Categories → Recipe List → Recipe Details → Add to Day → back to Planner**
- **Planner → Shopping List → back to Planner**
- **Settings accessible from tab/menu**

3) User Stories

Story Title: Browse Recipes by Category

User Story:
As a **home cook**, I want to **browse recipes by category** so that I can **easily find dishes that match what I'm in the mood for**.
Acceptance Criteria:
☐ Categories are displayed on the Browse screen (e.g., Chicken, Beef, Seafood, Dessert).
☐ Tapping a category shows a list/grid of recipes in that category.
☐ Each recipe card shows **title** and **thumbnail image**.
☐ User can scroll through all recipes in the category.
☐ A back action returns the user to the category list.

Story Title: Search Recipes by Name

User Story:
As a **user**, I want to **search recipes by name** so that I can **find specific recipes quickly**.
Acceptance Criteria:
☐ A search bar is visible on the Browse screen.
☐ Typing a keyword displays matching recipes.
☐ Clearing the search returns to the default list.
☐ A "No results found" message appears when there are no matches.
☐ Tapping a search result opens the Recipe Details screen.

Story Title: View Recipe Details

User Story:
As a **home cook**, I want to **view full recipe details** so that I can **prepare ingredients and follow steps correctly**.
Acceptance Criteria:
☐ Recipe Details displays the **recipe name** and **image**.
☐ Recipe Details displays **category** and **cuisine/area** (or "Unknown" if missing).
☐ Ingredients display as a list (ingredient + measure).

☐ Instructions display clearly (readable steps).
☐ If key fields are missing, the app shows a friendly fallback message.

Story Title: Add a Recipe to a Day on the Weekly Planner

User Story:
As a **weekly planner**, I want to **add a recipe to a specific day** so that I can **build a weekly meal plan**.
Acceptance Criteria:
☐ The Weekly Planner displays 7 days for the selected week.
☐ Each day shows "+ Add Meal" when empty.
☐ Selecting a recipe assigns it to the chosen day and updates the Planner.
☐ User can replace a planned meal with another recipe.
☐ Planned meals persist between app sessions (saved locally).

Story Title: Remove a Planned Meal from a Day

User Story:
As a **planner**, I want to **remove a meal from a day** so that I can **adjust my plan quickly**.
Acceptance Criteria:
☐ Planned days provide a remove/clear option.
☐ Clearing a day returns it to "+ Add Meal".
☐ Shopping List updates after a meal is removed.
☐ Changes persist between app sessions.
☐ Remove action is disabled/hidden when the day is already empty.

Story Title: Generate a Shopping List from Planned Meals

User Story:
As a **shopper**, I want to **generate a shopping list from my planned meals** so that I can **buy everything I need without forgetting ingredients**.
Acceptance Criteria:
☐ Shopping List includes ingredients from all meals planned for the selected week.
☐ Duplicate ingredients are merged into one item (by ingredient name).
☐ Each item shows ingredient name (and measure text if available).
☐ Each item has a checkbox to mark it purchased.
☐ If no meals are planned, show "No meals planned for this week."

Story Title: Favorite a Recipe

User Story:
As a **home cook**, I want to **mark/unmark a recipe as a favorite from the Recipe Details screen**, so that I can **remember recipes I like later without having to search again**.

Acceptance Criteria:
☐ A heart icon is visible on the Recipe Details screen.
☐ Tapping the heart toggles favorite status (on/off).
☐ Favorite status is saved locally and persists between app sessions.
☐ When a favorited recipe is opened again, the heart shows as filled/on.
☐ User can unfavorite a recipe by tapping the heart again.

Story Title: Set User Preferences (Dark Mode + Excluded Categories)

User Story:
As a **user**, I want to **set preferences (dark mode and excluded categories)** so that the app **fits my comfort and avoids unwanted recipes**.
Acceptance Criteria:
☐ A Settings screen is accessible in the app.
☐ User can choose Dark Mode setting (On/Off/System).
☐ User can exclude at least one category (example: Beef).
☐ Excluded categories do not appear in Browse results while enabled.
☐ Preferences persist between app sessions (saved locally).

Story Title: Handle Network Errors and Retry

User Story:
As a **user**, I want the app to **show clear error messages and allow retry** so that I can **recover when the internet or API fails**.
Acceptance Criteria:
☐ A loading indicator is shown while fetching recipes.
☐ If a request fails, a clear error message is displayed.
☐ A Retry option re-attempts the failed request.
☐ The app does not crash when the API returns missing/blank fields.
☐ “No results found” is shown for valid empty results.